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under Grant Agreement no. 101102611

T1.6.4 Potential application of the EUDI Wallet on EWP Services

Including Online Learning Agreement (OLA) and Erasmus+ App

Revision: v.1.0

Work package	WP 5
Task	1.6.4
Due date	dd/mm/yyyy
Submission date	16/01/2024
Deliverable lead	European University Foundation – Campus Europae (EUF-CE), Aristotle University of Thessaloniki (AUTh), Eötvös Lórand University (ELTE)
Version	1.0
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Abstract	This report outlines the impact the use of a digital Wallet can have on the implementation of specific services within the EWP network.
Keywords	EUDI Wallet, EWP Services, Learning Agreement, Erasmus+ App, trust chain

Document Revision History

Version	Date	Description of change	List of contributor(s)
V1.0	15/01/2025	1st version of the deliverable	Tiago Simões, Argyris Mpesinas, Péter Toberling, János Liszka, Helena Alves

Project co-funded by the European Commission in the Digital Europe Programme		
Nature of the deliverable:	**R ***	
Dissemination Level		
PU	Public, fully open, e.g. web	☐
CL	Classified, information as referred to in Commission Decision 2001/844/EC	
CO	Confidential to DC4EU project and Commission Services	

** R: Document, report (excluding the periodic and final reports)*

DEM: Demonstrator, pilot, prototype, plan designs

DEC: Websites, patents filing, press & media actions, videos, etc.

OTHER: Software, technical diagram, etc.

EXECUTIVE SUMMARY

The present report aims to review select Erasmus Without Paper (EWP) services in terms of their workflows and interactions involving citizen actors so as to assess potential applications of the EUDI Wallet in present or prospective processes.

The analysis of EWP services is extended to user-facing applications such as the Online Learning Agreement and the Erasmus+ App, which by their very nature are the most relevant points where citizen actors interact with the EWP network.

The analysis takes into account the different processes, actors and documents involved in the relevant services, as well as the presently defined features of the EUDI Wallet in order to assess the potential applications, feasibility and impact of such integration.

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ABBREVIATIONS

EWP	Erasmus Without Paper
EC	European Commission
API	Application Programming Interface
CNR	Change Notification Receiver
HEI	Higher Education Institution
IRO	International Relations Office
IIA	Interinstitutional Agreement
LA	Learning Agreement
OLA	Online Learning Agreement
ToR	Transcript of Records
E+App	Erasmus+ App

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1. TRUST MECHANISMS IN ERASMUS WITHOUT PAPER

Erasmus Without Paper (EWP), a part of the European Student Card Initiative, is "[a] digital solution for higher education institutions to connect their Erasmus+ mobility management systems so they can manage their mobility students online."[1]

EWP consists of a decentralized network where each participating Higher Education Institution (HEI) is a node; the network is only open to HEIs, and at present no other entities can be nodes (with exceptions made for the EC itself); every time a new node is added a KIC process takes place to ensure scope and access rights are rigorously maintained and implemented. By implementing APIs that conform to the API specifications as defined by entire technical community of users, HEIs can join the network and exchange documents with their partner institutions.

HEIs can join the EWP network via their own in-house systems, via commercial providers or via the Erasmus+ Dashboard[2]. The participation of a HEI in the EWP network requires providing information in the EWP Registry such as which APIs are implemented and what methods of secure communication are available to other nodes in the network.

EWP is designed to guarantee secure communication between nodes at various levels: client authentication, server authentication, request encryption and response encryption. For more detailed information see the EWP Authentication and Security repository on GitHub[3].

Due to its decentralized architecture, the EWP network is built on trust between connected systems based on the trust mechanisms described above.

It is important to note that the access of a HEI representative to their respective EWP node, which allows the representative to interact with EWP services, is governed by the processes put in place at the institutional level and/or at the third-party provider level.

The interaction between a HEI representative and their institutional systems connected to the EWP network is not governed by rules and procedures related to the EWP services in question.

[1] https://erasmus-plus.ec.europa.eu/european-student-card-initiative/ewp

[2] https://erasmus-plus.ec.europa.eu/european-student-card-initiative/ewp/dashboard

[3] https://github.com/erasmus-without-paper/ewp-specs-sec-intro

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2. ERASMUS WITHOUT PAPER SERVICES

The select EWP services under analysis encompass all the required steps for student mobility to take place. These are, in chronological order within the mobility process, the following:

- **Inter-institutional Agreement (IIA)**
  The IIA is a document signed between two HEIs where the respective conditions of cooperation are outlined. This is the basis for mobility to occur between the two HEIs.

- **Nomination for mobility**
  In the context of EWP, a nomination is the process whereby a sending HEI submits an Outgoing Mobility associated with a student to a receiving HEI.

- **Learning Agreement (LA)**
  The LA is a document signed between three parties: a student who wishes to go on mobility, the sending HEI (or home institution) which the student attends and the receiving HEI (or host institution) where the student aims to spend their mobility period.

- **Transcript of Records (ToR)**
  The ToR is a document that outlines what a student has accomplished during their mobility period at the receiving HEI, or host institution. The information contained therein is transmitted to the sending HEI, or home institution, for purposes of credit recognition.

A more detailed overview of the student mobility business process is available on GitHub[4]. The relevant flowcharts from this repository are reproduced in the current report for clarity.

[4] https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts/

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2.1. INTERINSTITUTIONAL AGREEMENT (IIA) WORKFLOW

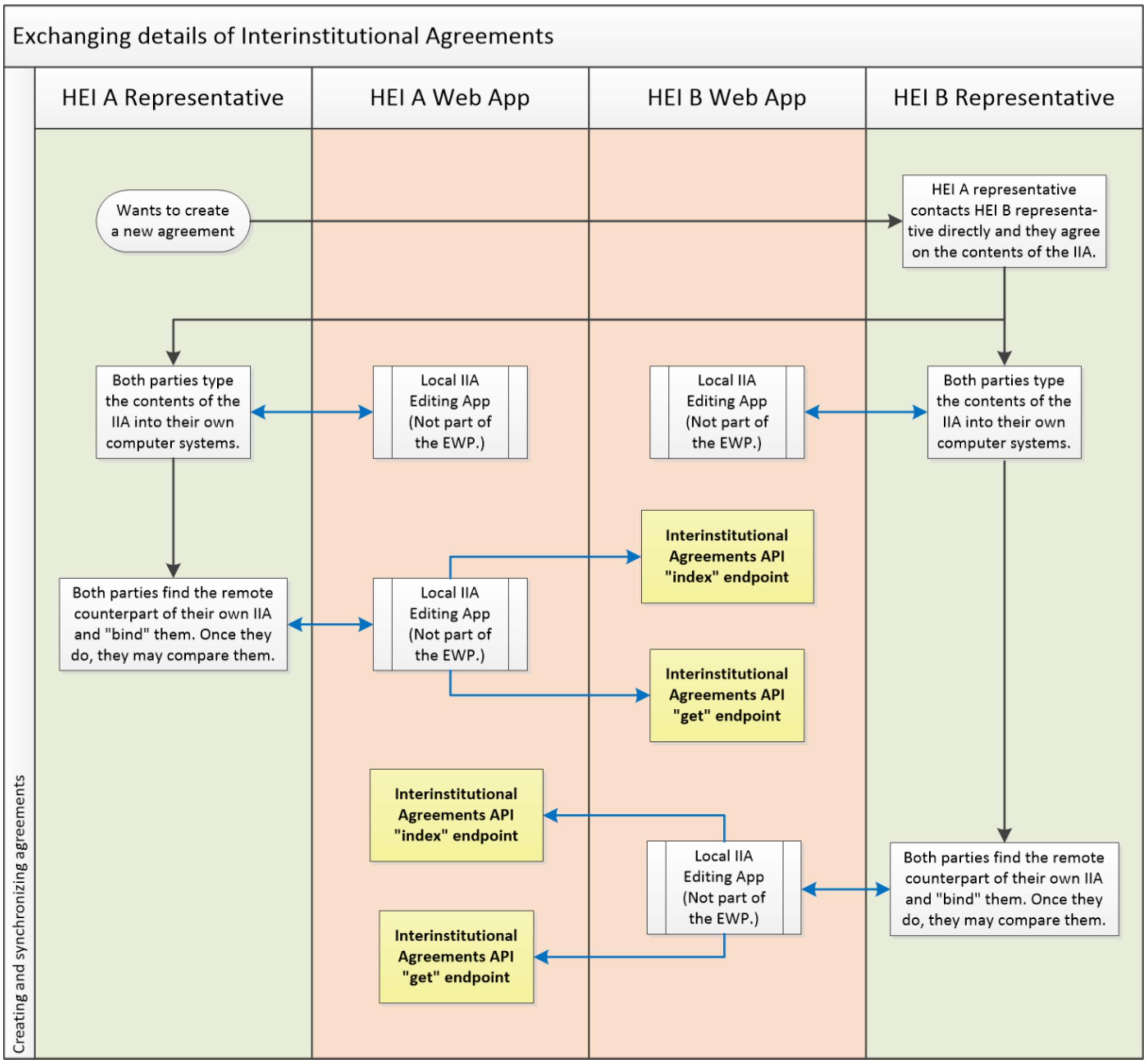

Figure 1 - Process of exchanging interinstitutional agreements. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

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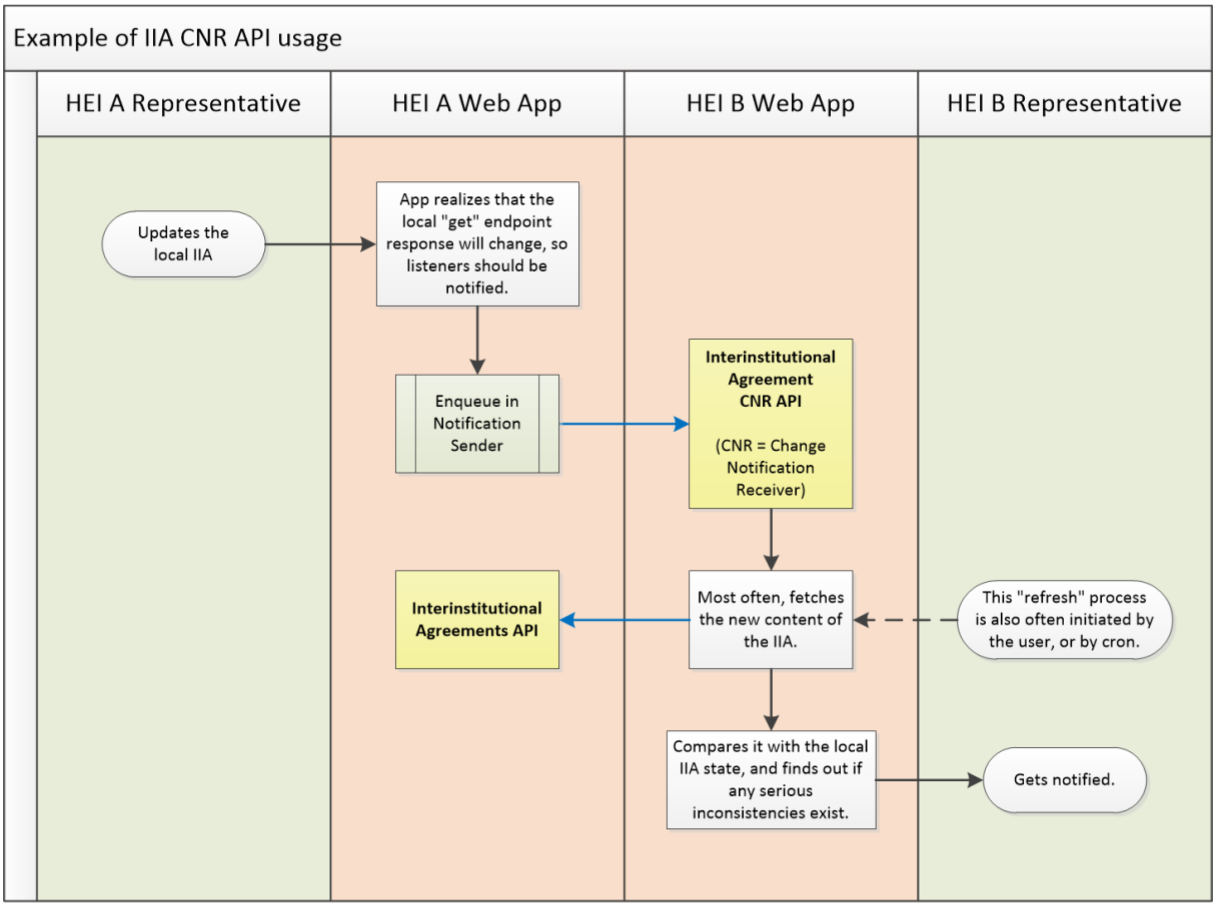

Figure 2 - Example of IIA CNR API usage. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

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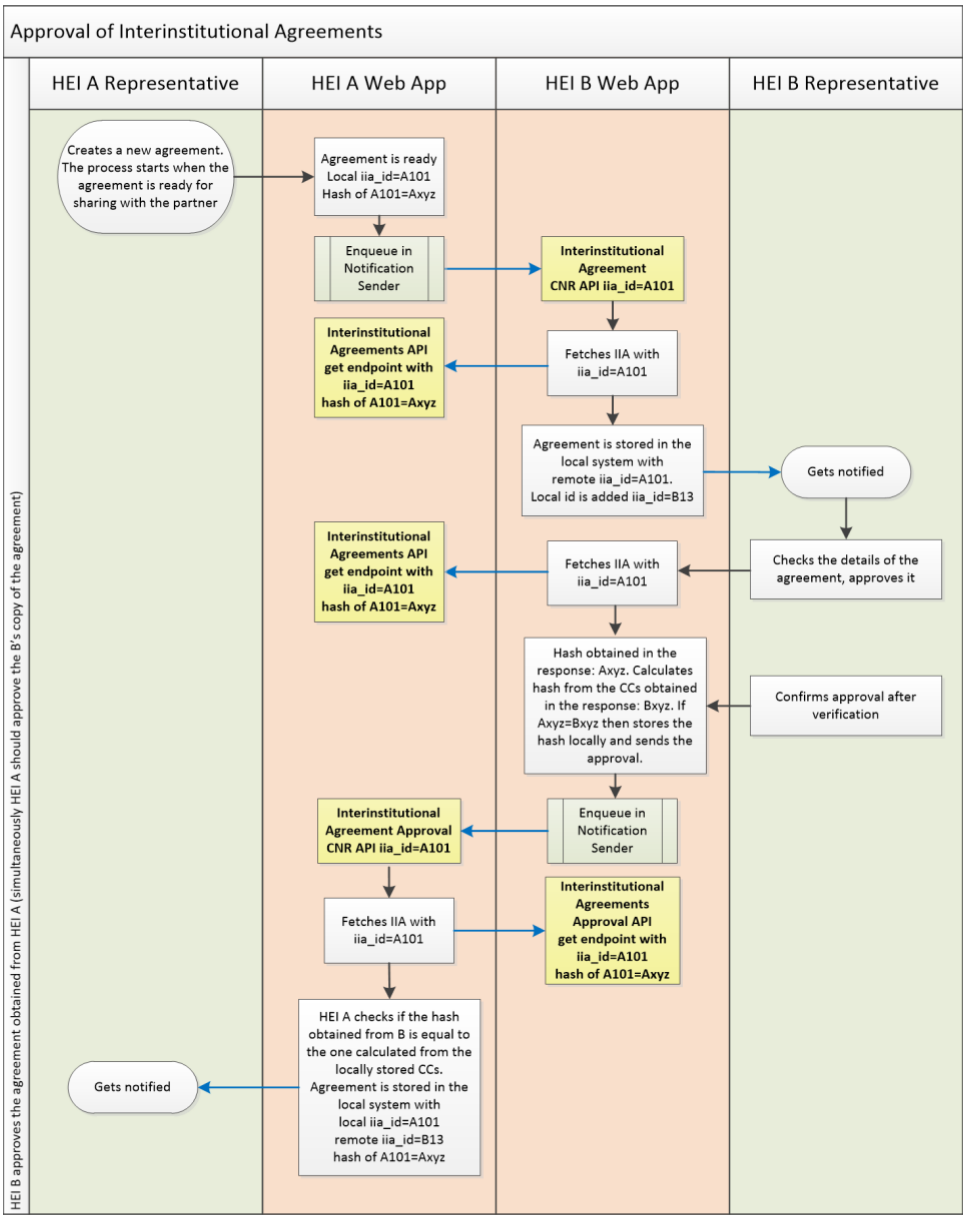

Figure 3 - Process of approval of Interinstitutional Agreements. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

As is clear from the diagrams in figures 1 through 3, the IIA is created and approved by representatives of each HEI. The IIA, however, is not a document that pertains to each HEI representative as a citizen actor. In fact, it is possible that the representative of a given HEI changes during the process.

Since there are no documents directly pertaining to citizen actors, there is no use case for a digital wallet.

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2.2. NOMINATIONS FOR MOBILITY

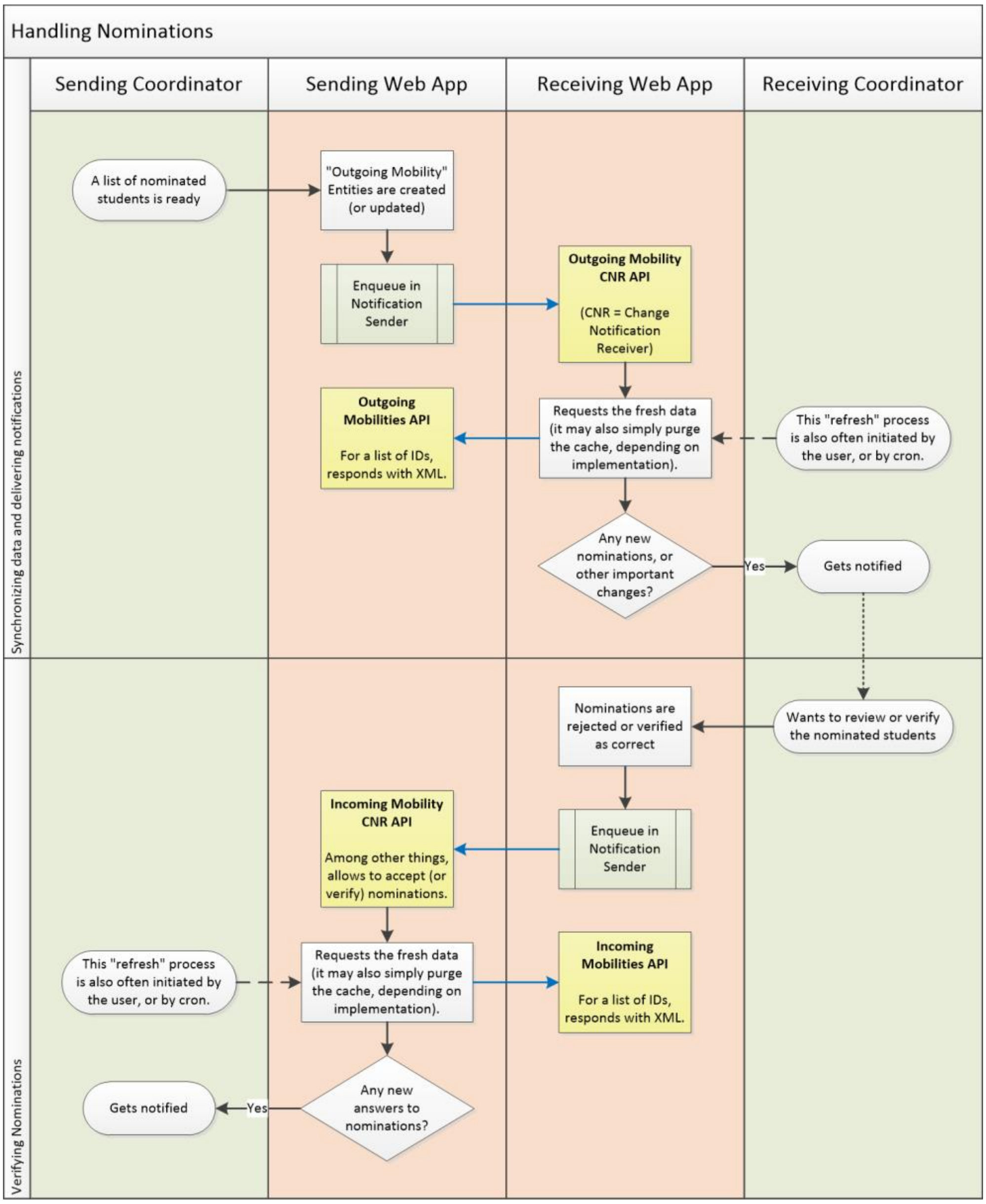

Figure 4 - Process of handling nominations. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

The nomination process involves a coordinator of a student's home institution, who submits an outgoing mobility application associated with the student, and a coordinator of a host institution who reviews and either approves or rejects the nomination. When considering the coordinators as citizen actors, since there are no documents directly pertaining to them, there is no use case for a digital wallet.

The nomination process, however, may also require the student as a citizen actor to provide documentation to be attached to the nomination. As such the student's interactions can be analysed through the prism of the most widely available interface for students: The Erasmus+ App.

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2.2.1. NOMINATIONS IN PRACTICE: THE ERASMUS+ APP

The Erasmus+ App is a student-facing application that provides useful information and several tools to assist users before and during mobility. By design, many features of the Erasmus+ App are accessible for users without logging, and sharing personal data or credentials are limited to a few selected cases where these will be used to evaluate a user's eligibility for mobility. As a result, this also limits the number of potential applications of a EUDI Wallet in relation with the E+ App, effectively narrowing it down to the mobility application process.

Potential use cases

1. **Logging in via Wallet**

Creation of accounts and login is done through external authentication services (EU Login and MyAcademicID). Though the current login methods provide sufficient information (especially when done via MyAcademicID-eduGAIN), relying on the Wallet to log in and authenticate users is an alternative to consider. This possibility should be examined later when there are more detailed outputs available in the project to see if these can be adapted.

2. **Destination selection and applying for mobility**

One of the core features of the E+ App is the possibility for a student to apply for mobility via the App. This possibility is only available if their home university has decided to process applications via the Erasmus+ App and uses a compatible mobility management software to allow data transfer via the standard API.

In practice there are two features linked to the complete user process flow: the first feature is the destination finder that allows students to browse through the possible hosts, saving and selecting them to apply for later. The other feature is the application itself, where the user may upload all the necessary documents and information required by both the home and host universities then send them to the home institution for evaluation.

In this process some possible application of the Wallet can be identified, though as mentioned before, its use may be limited at this point.

a. Profile information

As a preliminary step before being able to access the destination finder, the user needs to fill out their profile, including the academic information. These data will be then used by the feature to filter the possible matching destinations. The profile data contains the following range of information:

1. Erasmus programme
2. Home Institution
3. Host institution (optional, irrelevant for this feature)
4. Academic profile information
    a. Faculty at home institution
    b. Study level

c. Study field

d. Number of ECTS credits completed

e. Language skills

Assuming any of these data – most notably language skills in the form of credentials – are stored in the Wallet, fetching these from the Wallet might be possible. However, this feature needs to provide a certain level of flexibility to users. That means that they are free to provide and modify these data as they see fit – even assuming a later time in the future when they have already acquired certain qualifications – to customize their searching parameters. If we would link this to an automatic data fetching that may enforce or override the user provided data with their actual status at any time, it may be unnecessarily constraining, therefore the cost of implementation may easily exceed the potential gain of prefilling a limited number of fields.

b. Destination finder

Users may browse through the potential destinations (host universities and their data sheets) based on their profile and may save any to apply for later. There is no additional data requested or used here other than the ones listed in Appendix I.

c. Applying for mobility

Once the user has selected at least one destination, they may initiate the application feature. To apply for mobility the user needs to provide a range of documentations and credentials, based on two sets of requirements: 1. The required documentation determined by their home institution, and 2. The required documentation determined by the selected host institution(s). Currently these documents are uploaded and sent to the home institution through an API. Note: resharing with the host is the task of the home institution (nomination process) done independently from the E+ App.

From the perspective of the potential applicability of the EUDIW, there are basically two types of documentations provided here:

- Type1: documentations where providing the credentials would be eligible (e.g. language certificate, diploma)
- Type2: documentations where accessing the whole content is necessary (e.g. CVs, motivation letters, portfolios)

Since the current implementation of the Wallet aims at storing and managing digital credentials, but not at granting access to the whole content of any type of documentations, Type2 remains out of scope.

In case of Type1 relying on the wallet could be beneficial. In the current practice digital copies of certificates are uploaded and shared, and the verification of these documentations remains the task of the IROs at the home universities (and the same documents are sent and shared with the potential host institutions in the follow-up process-step, that is the nomination). Would this step be replaced by the option to share the credentials stored in EUDIW and verified by trusted partners already, it could eliminate the need to create, upload and store digital copies of educational credentials as well as sending and potentially multiplying documentations containing personal information. On the technical side this would require the following:

1. Create the ability to link the EUDIW of a user to their E+ App account (may be skipped if implemented in a way that users may provide external links to their digital credentials)

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2. Modifying the application frontend and backend to allow users to add and save links to their digital credentials stored in the EUDIW (note: keeping the original upload functionality will be necessary until the EUDIW can become a widespread standard solution for managing credentials)
3. Adjust the APIs to allow sending the links to the digital credentials (note: indirectly this may also affect the nomination API, that allows home and host institutions to share the applicants' data)
4. Allowing institutional users (IROs) to check digital credentials provided by the applicants (out-of-scope for the E+ App, depends on the mobility management software implementation of the home – and potentially host – institutions)

The actual use cases and optimal ways to implement in the E+ App should be determined at a later stage, based on the outputs of the ongoing design and testing activities of the project.

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2.3. LEARNING AGREEMENT (LA)

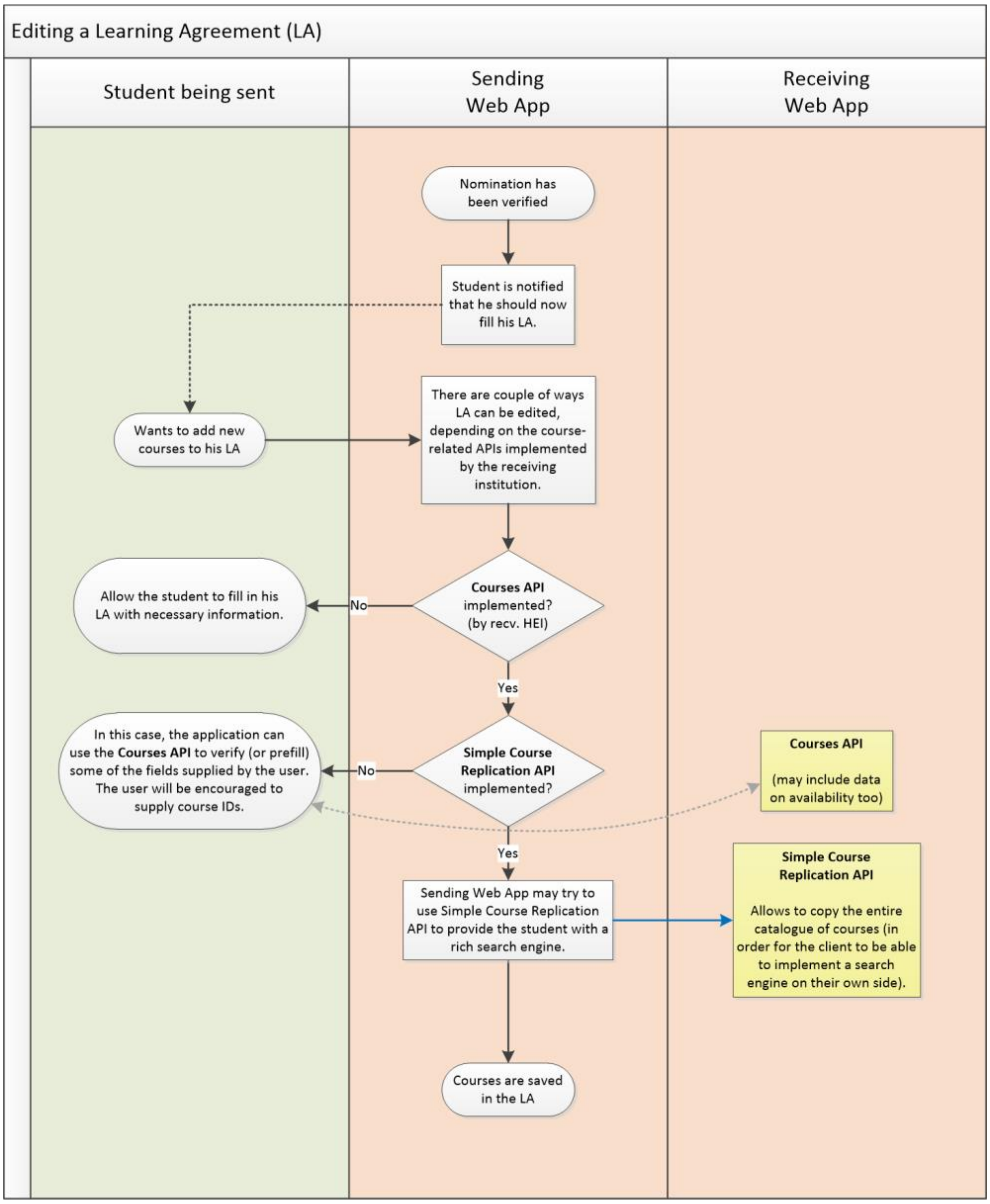

Figure 5 - Process of editing a learning agreement. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

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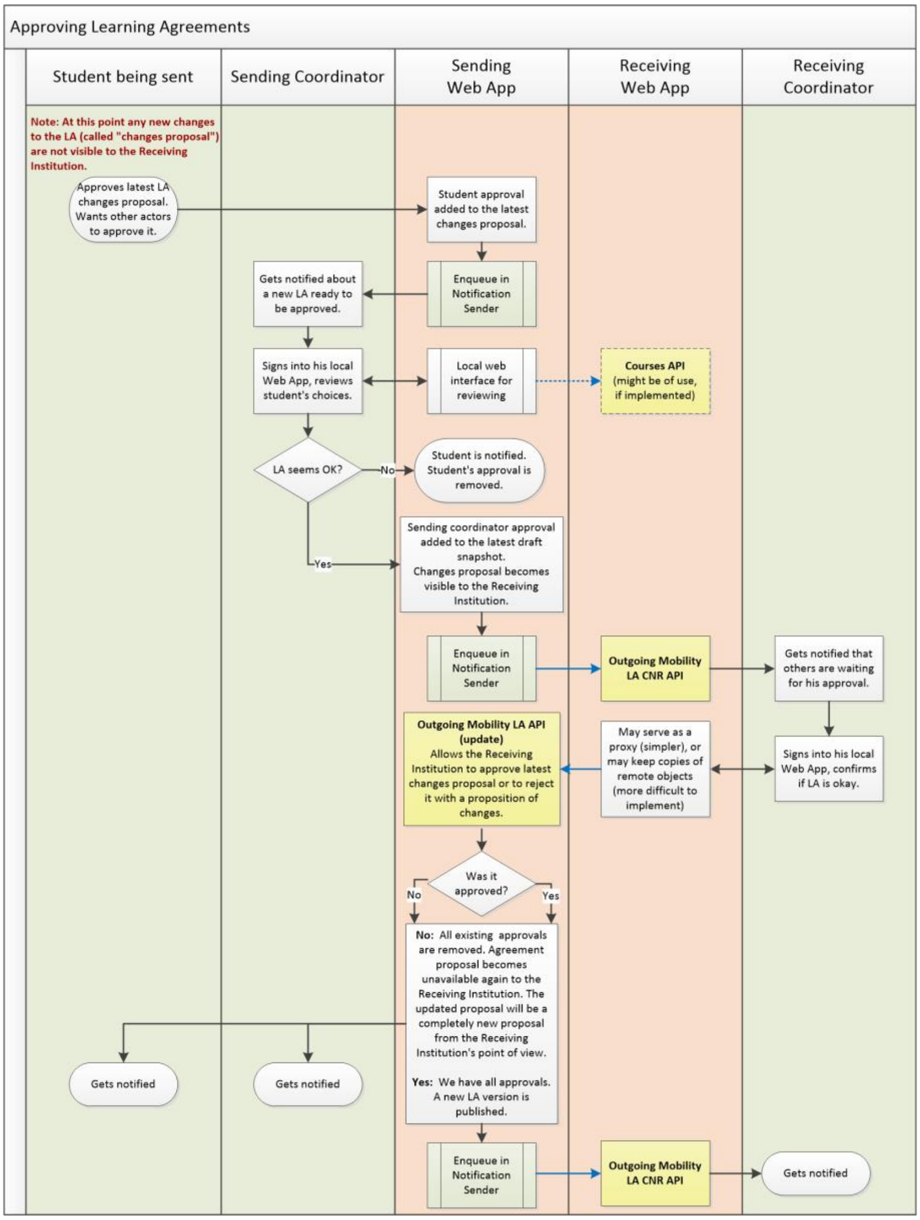

Figure 6 - Process of approving Learning Agreements. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

As can be seen in the diagrams in Figures 5 and 6, there are two institutional actors – sending and receiving HEIs – and one citizen actor involved – the student. The LA does not constitute documentation pertaining to the HEI representatives as citizen actors, which means that there is no use case for a digital wallet at that level.

The student, however, is at the centre of the process, as a citizen actor, and so their interaction in this process merits further investigation into potential applications of the EUDI Wallet.

Since the most relevant interface for students to manage their LAs is the Online Learning Agreement platform (OLA), a dedicated analysis has been conducted by the maintainers of the OLA.

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2.3.1. LEARNING AGREEMENT IN PRACTICE: ONLINE LEARNING AGREEMENT (OLA)

For information about the data fields contained in a learning agreement, see APPENDIX I.

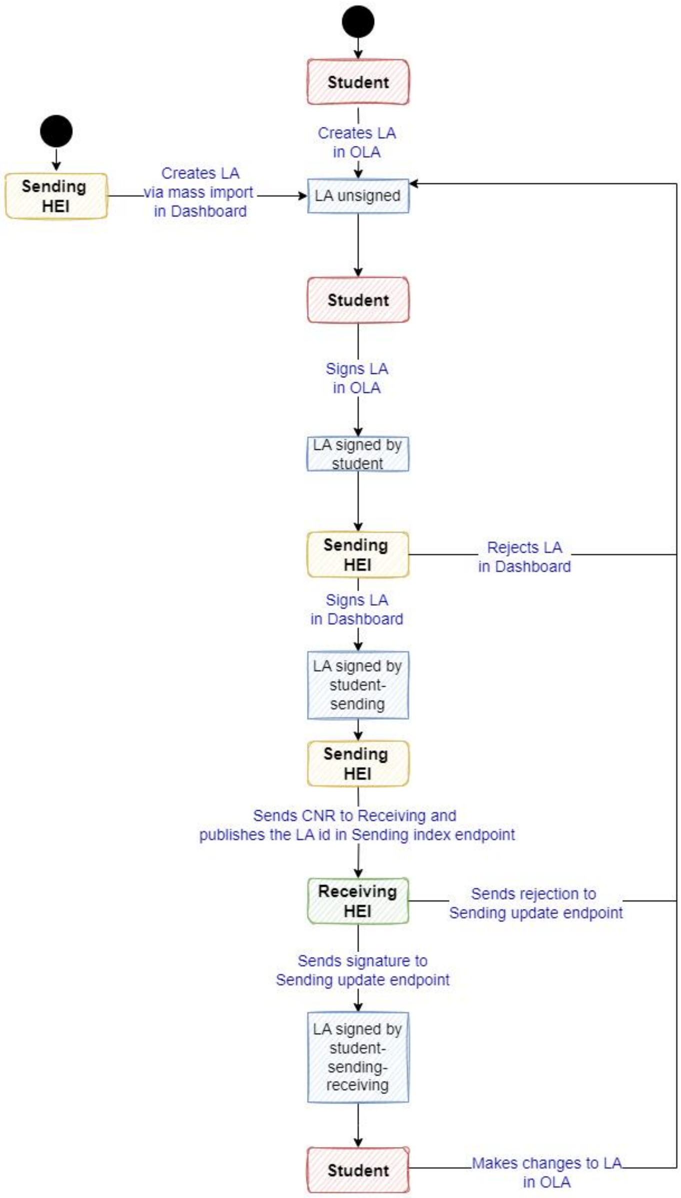

Figure 7 - LA workflow involving the OLA and the Erasmus+ Dashboard

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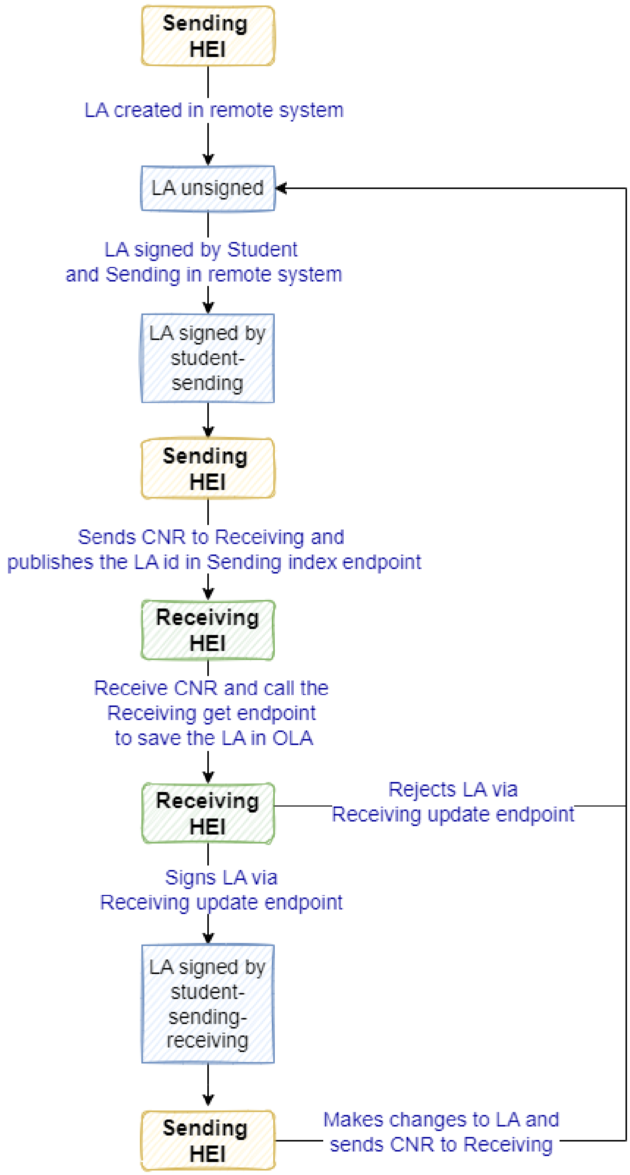

Figure 8 - LA workflow involving remote systems

There are three types of agreements in OLA:

- Semester mobility
- Blended mobility with short-term physical mobility
- Short-term Doctoral mobility

The Semester mobility consists of all the data mentioned above and its lifecycle follows the provided flow diagrams. After the agreement has been signed by all the concerned parties (Student, Sending HEI IRO, Receiving HEI IRO) the student can at any point make changes to the agreement, which restart the signing process. There is no limit to how many times this can occur during the lifetime of a learning agreement. Semester mobilities make up the majority of created learning agreements.

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In contrast Blended and doctoral mobility learning agreements contain all the Student, HEI, general and signature fields but have only table C. Their flow is also finished after it has been signed by all of the concerned parties (Student, Sending HEI IRO, Receiving HEI IRO). The student cannot make further changes to the agreement.

It is also important to note that the significance of the learning agreement in the context of an Erasmus Mobility pales in comparison to that of the transcript of records. The transcript of records contains the final list of the courses the transfer student studied at the receiving institution, along with the grades they received. It also contains the final specific dates regarding the duration of the mobility, which corresponds to the grant received by the student for the Erasmus mobility.

The learning agreement is a document that by definition is shared to all relevant involved parties. We do not foresee a case where a need arises to share the learning agreement with any third parties outside the mobility officers of the two institutions and the student and all those parties had a part in the creation and approval of the agreement.

For the above reasons, combined with the mutability of the learning agreement and lack of a final immutable state, we are of the opinion that learning agreements do not make a good target for inclusion in the digital wallet.

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2.4. TRANSCRIPT OF RECORDS (TOR)

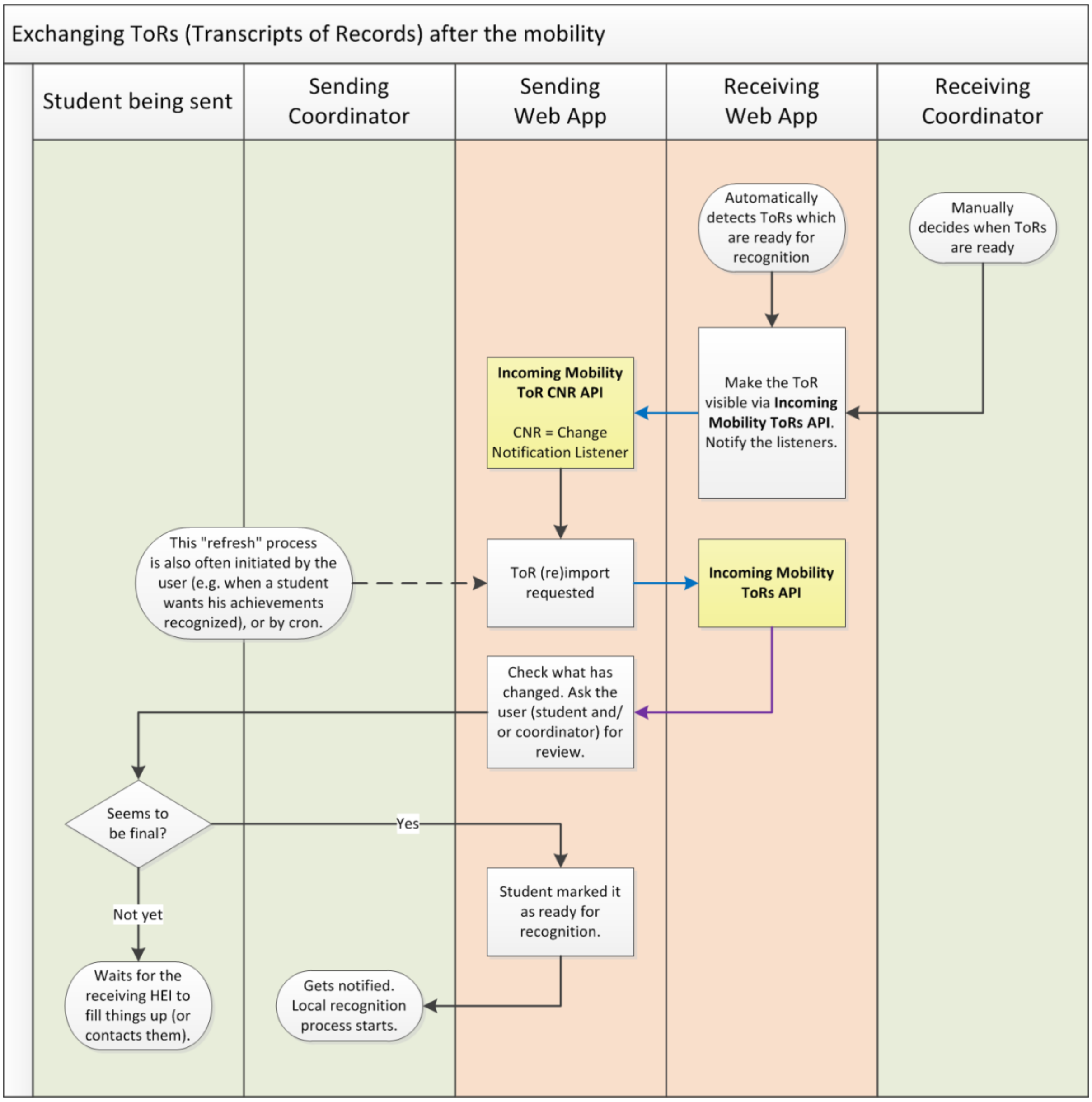

Figure 9 - Exchanging ToRs after the mobility. Retrieved from https://github.com/erasmus-without-paper/ewp-specs-mobility-flowcharts

In the diagram above the listed parties are the same as in the case of the LA: a representative of the sending HEI, a representative of the receiving HEI and a student, who is the only citizen actor identified in the workflow.

Considering the role of the student as described in this workflow, it becomes clear that the only involvement of the student is in monitoring the completion of the ToR, while not effectively taking possession of the document as such.

The relevant information contained in the ToR will later be used for credit recognition and will appear in the diploma supplement, at which point it is worth considering the potential role of the EUDI Wallet. However, that use case is out of the scope of EWP.

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In conclusion, no potential application of the EUDI Wallet has been identified in relation to the ToR as it exists at present. However, should the role of the ToR substantially change into a standalone role which would directly involve the student as a citizen actor, this issue should be reevaluated in light of the features offered by the EUDI Wallet.

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CONCLUSION

This report aimed to evaluate the potential application of the European Digital Identity (EUDI) Wallet within EWP services and to understand the impact the widespread adoption of the EUDI Wallet would have on its implementation.

There are limited immediate use cases for processes such as Inter Institutional Agreements and nominations for mobility, as these largely involve institutional actors rather than individual citizens. However, for students, integrating digital credentials such as language certificates into the Erasmus+ App offers an opportunity to streamline workflows and enhance security.

In the specific case of Learning Agreements (LAs), their mutable nature and reliance on institutional actors reduce their compatibility with the EUDI Wallet, although some potential exists for student-facing applications. Similarly, while Transcripts of Records (ToRs) currently involve minimal direct interaction with students, future changes to their role could create opportunities for wallet integration.

Moving forward, the integration of digital credentials in the Erasmus+ App could be piloted within the project. Additionally, monitoring the evolving role of documents like ToRs will help future-proof the applications of the wallet. These steps would enable the project to advance digital transformation in European student mobility, leveraging the EUDI Wallet to enhance the efficiency of the ecosystem.

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APPENDIX A - LEARNING AGREEMENT DATA FIELDS

LA Data fields

Student data

- First name(s)
- Last name(s)
- Email
- Date of birth
- Gender
- Nationality
- Field of Education
- Field of education
- Field of Education Comment
- Study cycle

Sending and receiving HEI information:

- Country
- HEI Name
- HEI ID
- Erasmus code
- Faculty/Department

Responsible person/ Contact person data

- First name(s)
- Last name(s)
- Position
- Email
- Phone number

General LA data fields

- Academic year
- Planned start of the mobility
- Planned end of the mobility
- The main language of instruction at the Receiving Institution
- The level of language competence
- Web link to the course catalogue at the Receiving Institution describing the learning outcome
- Provisions applying if the student does not complete successfully some educational components

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- Web link to the course catalogue at the Sending Institution describing the learning outcomes

Table A - Study programme at the Receiving institution (contains multiple components)

- Component title
- Component Code
- Number of ECTS credits
- Semester

Table B - Recognition at the Sending institution (contains multiple components)

- Component title
- Component Code
- Number of ECTS credits
- Semester
- Automatically recognised towards student degree
- Automatic recognition comment

Table C - virtual components

- Component title or description at the Receiving Institution
- Component Code
- Number of ECTS credits (or equivalent) to be recognised by the Sending Institution
- Short description of the virtual component
- Automatically recognised towards student degree
- Automatic recognition comment

SIgnatures

- Student signature
- Sending HEI IRO signature
- Receiving HEI IRO signature

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